Linear Discriminant Analysis (LDA) – MCQs

1. What is the primary objective of Linear Discriminant Analysis (LDA)?
    - A) Minimize reconstruction error
    - B) Maximize class separability
    - C) Minimize classification accuracy
    - D) Increase dimensionality
2. LDA is primarily used for:
    - A) Clustering
    - B) Regression
    - C) Dimensionality Reduction and Classification
    - D) Data Augmentation
3. Which matrices are central to the formulation of LDA?
    - A) Covariance and Correlation Matrices
    - B) Within-Class Scatter and Between-Class Scatter Matrices
    - C) Adjacency and Incidence Matrices
    - D) Kernel and Gram Matrices
4. The within-class scatter matrix represents:
    - A) Separation between class means
    - B) Variance of samples within each class
    - C) Distance between clusters
    - D) Prediction error
5. The between-class scatter matrix measures:
    - A) Variability within classes
    - B) Separation among class means
    - C) Noise variance
    - D) Reconstruction loss
6. LDA seeks projection directions that:
    - A) Minimize both within-class and between-class variance
    - B) Maximize within-class variance
    - C) Maximize between-class variance while minimizing within-class variance

- D) Ignore class labels

7. For a dataset with C classes, the maximum number of LDA components is:
   - A) C
   - B) C + 1
   - C) C − 1
   - D) Equal to the number of features
8. Which of the following is an assumption of classical LDA?
   - A) Classes have identical covariance matrices
   - B) Classes have different covariance matrices
   - C) Features are independent and binary
   - D) Data must be non-linear
9. LDA is a supervised technique because:
   - A) It uses unlabeled data
   - B) It requires class labels
   - C) It uses reinforcement signals
   - D) It performs clustering
10. In LDA, eigenvectors are obtained from:
    - A) Covariance Matrix
    - B) Correlation Matrix
    - C) $S_W^{-1} S_B$
    - D) Gram Matrix
11. Compared to PCA, LDA:
    - A) Does not reduce dimensions
    - B) Ignores class labels
    - C) Uses class information
    - D) Requires more dimensions
12. Which criterion is maximized in Fisher's Linear Discriminant?
    - A) Reconstruction Error
    - B) Variance Ratio
    - C) Between-Class Scatter to Within-Class Scatter Ratio
    - D) Mean Squared Error
13. LDA assumes data from each class follows:

- A) Uniform Distribution
- B) Poisson Distribution
- C) Gaussian Distribution
- D) Exponential Distribution

14. If there are two classes, the maximum number of discriminant components is:
    - A) 0
    - B) 1
    - C) 2
    - D) Number of features
15. Which statement best differentiates PCA and LDA?
    - A) PCA is supervised; LDA is unsupervised
    - B) PCA maximizes variance; LDA maximizes class separability
    - C) PCA requires labels; LDA does not
    - D) Both optimize the same objective
16. What happens if class labels are randomly shuffled before applying LDA?
    - A) Performance generally degrades
    - B) Performance improves
    - C) No effect
    - D) Number of features increases
17. Which matrix must generally be invertible in standard LDA?
    - A) Between-Class Scatter Matrix
    - B) Within-Class Scatter Matrix
    - C) Correlation Matrix
    - D) Distance Matrix
18. LDA projects data onto:
    - A) Directions of maximum variance
    - B) Random directions
    - C) Directions maximizing discrimination among classes
    - D) Kernel feature space only
19. In pattern recognition, LDA is often applied:
    - A) Before classification
    - B) After classification

- C) Only during testing
- D) Only for clustering

20. Which of the following can be a limitation of LDA?
    - A) Requires labeled data
    - B) Assumes linear boundaries
    - C) Sensitive to covariance estimation
    - D) All of the above
21. LDA can be viewed as:
    - A) Feature Extraction Technique
    - B) Data Compression Technique
    - C) Classification Technique
    - D) All of the above
22. The rank of the between-class scatter matrix is at most:
    - A) Number of Features
    - B) Number of Samples
    - C) Number of Classes Minus One
    - D) Twice the Number of Classes
23. Which application commonly uses LDA?
    - A) Face Recognition
    - B) Speech Recognition
    - C) Medical Diagnosis
    - D) All of the above
24. In Fisher's criterion, a good projection should:
    - A) Bring class means closer together
    - B) Increase overlap among classes
    - C) Separate class means and reduce class spread
    - D) Ignore variance
25. LDA is most effective when:
    - A) Classes are well separated
    - B) Data is completely random
    - C) No labels are available
    - D) Features are constant

26. Which of the following affects LDA performance?
    - A) Class imbalance
    - B) Outliers
    - C) Small sample size
    - D) All of the above
27. The generalized eigenvalue problem in LDA is solved to obtain:
    - A) Cluster Centroids
    - B) Principal Components
    - C) Discriminant Directions
    - D) Decision Trees
28. LDA is likely to perform poorly when:
    - A) Classes are linearly separable
    - B) Covariance assumptions are heavily violated
    - C) Labels are accurate
    - D) Features are informative
29. Which quantity is minimized in the denominator of Fisher's criterion?
    - A) Between-Class Scatter
    - B) Within-Class Scatter
    - C) Total Variance
    - D) Correlation
30. In multiclass classification, LDA creates:
    - A) One discriminant axis only
    - B) Multiple discriminant axes up to $C - 1$
    - C) One axis per feature
    - D) No projection axes
31. LDA is particularly useful when:
    - A) Number of features is very small
    - B) Feature reduction is needed before classification
    - C) Data contains no labels
    - D) Only clustering is required
32. Which assumption is shared by many LDA formulations?
    - A) Equal class priors

- B) Equal covariance matrices across classes
- C) Independent samples only
- D) Nonlinear decision surfaces

33. The output of LDA is typically:
    - A) Cluster Assignments
    - B) Reduced-Dimensional Feature Vectors
    - C) Regression Coefficients
    - D) Association Rules
34. Why is LDA often preferred over PCA for classification tasks?
    - A) It ignores variance
    - B) It uses label information
    - C) It requires fewer samples
    - D) It is unsupervised
35. If there are 5 classes in a dataset, the maximum number of discriminant vectors produced by LDA is:
    - A) 3
    - B) 4
    - C) 5
    - D) 6
36. Which type of learning does LDA belong to?
    - A) Supervised Learning
    - B) Unsupervised Learning
    - C) Reinforcement Learning
    - D) Self-Supervised Learning
37. What role do class means play in LDA?
    - A) No role
    - B) Used to compute between-class scatter
    - C) Used only for normalization
    - D) Used only during testing
38. Which of the following is true about LDA decision boundaries?
    - A) Always nonlinear
    - B) Typically linear
    - C) Always quadratic

- D) Always circular

39. LDA can suffer from the "small sample size" problem when:
    - A) Number of samples is much smaller than number of features
    - B) Number of features is zero
    - C) Classes are balanced
    - D) Data is normalized
40. Which statement about LDA is correct?
    - A) It ignores class membership
    - B) It maximizes variance without labels
    - C) It seeks projections that improve class discrimination
    - D) It is purely a clustering algorithm